



Examiners' Report

Summer 2026

Pearson Edexcel GCE

In Further Mathematics (9FM0)

Paper 4D Decision Mathematics 2

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General Comments

This paper allowed learners to demonstrate their understanding across a broad range of the specification, while also providing opportunities for differentiation. Many learners showed sound knowledge of the content and produced well-presented solutions using appropriate techniques, tables and diagrams. Some responses were more difficult to follow, and learners should be reminded that Decision Mathematics is a methods-based examination in which clear presentation of methods is important. Questions 4, 5 and 8 provided opportunities for learners to demonstrate more developed understanding and problem-solving skills.

Learners demonstrated improved understanding of the newer topics on the specification, particularly decision trees and recurrence relations. Responses to these topics were often set out with greater confidence and accuracy. For decision trees, learners should continue to focus on presenting solutions in the required format, including the use of appropriate node shapes, clear labelling and working values shown in the relevant nodes.

Comments on individual questions

Question 1

This question was generally well answered, with many responses showing a clear understanding of how to modify the initial table.

Some responses did not include all of the required modifications to the initial table. Common omissions included not adding the additional dummy column, not placing large values in the impossible cells, or not making either of these changes before beginning the row and column reductions. In some responses, row or column reductions were carried out before the initial table had been fully modified, with the impossible cells not treated appropriately. Learners should be reminded that the initial table must be adjusted fully before the reduction process begins to gain full marks.

A small number of responses treated the problem as a maximisation problem by subtracting the entries from a large number. This approach was not appropriate for the context of the question and limited the progress that could be credited.

Question 2

Many responses showed a clear understanding of the required processes and demonstrated the methods needed for this question effectively.

Part (a)

Part (a) required learners to demonstrate and provide an argument for domination. Almost all learners gained the first B mark for stating that Q dominated P (or an acceptable equivalent statement), however not all were able to justify this by providing the required inequalities.

Part (b)

Part (b) was well-answered, with almost all learners giving the correct table with rows and columns inverted and signs swapped. A small number of responses reversed the signs without swapping rows and columns; this was condoned on this occasion, but learners should be aware of the convention that the rows usually reflect the options of the player the game is focused on (Y, in this case). A small number of responses incorrectly removed option Q instead of option P;

this not only did not gain this mark but also affected a significant proportion of the remaining marks in the question.

Part (c)

Part (c) was a very standard question, and most learners answered it well scoring well. Some learners did not gain the first B mark for not fully defining probability; it is an expectation that learners fully define p and $(1 - p)$ as probabilities for their two options, and we are no longer condoning just writing p and $(1 - p)$ next to the relevant rows. Most responses were successful in forming the correct expressions, drawing acceptable graphs, and finding the correct value of p (along with the correct optimal strategy for Y). A small number of responses attempted variations (such as using the initial tableau with option P crossed out, and writing the negated expressions from this table, equivalent to the main scheme).

Part (d)

Part (d) was generally answered successfully by almost all learners who had reached a correct value of p in the previous part.

Question 3

This question provided an opportunity for differentiation, with responses showing a range of understanding of the required methods. Some responses demonstrated clear and accurate application of the relevant processes, while others showed that further attention is needed to the detail required in setting out complete solutions.

Part (a)

This part was usually answered well, and the responses who reached a value of 69 made progress to next parts.

Part (b)

Most responses gave the initial flow correctly. A majority also showed some correct understanding of the flow into and out of H, although in some cases the explanation or calculation was not complete.

Many responses to part (b) included all of the relevant information and showed a clear understanding of the method required. Learners should be reminded that,

in this methods-based paper, explanations and justifications need to include sufficient detail and should be supported by calculations where appropriate.

Part (c)

Part (c) proved challenging for many learners. Responses often showed difficulty in accounting for arcs travelling from sink to source through the cut, as well as in using the different values for flow, minimum capacity and maximum capacity on each arc. More successful responses identified the relevant arcs carefully and used the appropriate values to determine the cut correctly.

Part (d)

Part (d) also proved challenging. Some responses did not attempt this part, while others gave routes that could not be used as valid flow-augmenting routes. Some responses tried to list as a series of arcs (which was problematic given that the flow-augmenting route contained a backflow), whilst some responses omitted the requirement in the question to give the amount that the route was being augmented by. (Due to the presence of a second flow-augmenting route using SADFGT, this was required and learners did not gain the mark if they failed to give the additional flow increase of 4.)

Part (e)

Part (e) required learners to justify that the augmented flow was maximal by using the minimum-cut-maximum-flow theorem. Successful responses identified a minimum cut, stated the value of the cut and the value of the improved flow; then used the equality of these values to draw their conclusion that since there is a cut with equal value to the augmented flow, then by the minimum-cut-maximum-flow theorem, a maximal flow has been found. Some responses stated “max flow min cut” without providing sufficient supporting detail. Learners should be reminded that a complete justification needs to include the relevant cut, its value, the value of the improved flow, and a clear conclusion linked to the minimum-cut-maximum-flow theorem. The most common responses at present tend to score only the first mark for finding a cut; most typically a clear indication of the improved flow is not given and thus two of the three marks are not awarded.

Question 4

Many responses showed an understanding of the requirements of this question, although accuracy in carrying out the required methods was not always maintained throughout. This limited the extent to which some learners could complete the solution successfully.

Part (a)

Nearly all learners found an initial solution in part (a) correctly, and a majority recognised the need to include a zero (often citing the cell BR) for part (b).

Part (b)

However, part (b) also required that learners gave a brief explanation as to why the initial solution required modification; most learners gave an acceptable explanation, with many using the correct technical language of the initial solution being degenerate.

Part (c)

In part (c) students were required to perform one iteration of the stepping stone method to improve a given feasible solution. Most learners understood the requirement to find shadow costs and improvement indices, and many were able to complete this accurately to identify the correct entering cell for the second part of the algorithm. The particular challenge with this question was the possibility of exiting via two distinct cells; learners needed to choose one or the other. Many learners seemed either unaware of, or unprepared for, this requirement. Some learners gave both exiting cells, adding a zero to both cells and thus having an incorrect improved solution with 8 cells filled in, whilst others gave a degenerate solution as they did not know where to include the zero.

Part (d)

In part (d), some responses gave too many improvement indices, often following difficulties in identifying the appropriate improved solution in part (c). Successful responses applied the stepping stone method accurately and kept the working clearly linked to the feasible solution being tested. In some responses, slips in calculating shadow costs or improvement indices affected the accuracy of the final solution. Learners should be reminded to check that each improvement index is calculated from the correct occupied cells and that the final solution remains feasible.

Question 5

This question on a decision tree involving multiple decisions (the decision of which dance to perform as well as the decision as to whether to incorporate a lift) was an increase in challenge from trees topics. It was also examined more firmly; we feel that learners have now had sufficient time to understand the requirement to use the correct shaped node to represent a decision, chance or pay-off, and this was consistently not awarded marks where learners used the incorrect shape. It has also been made clear that the nodes should be large enough to contain the working EMVs throughout the problem, and that rounding should be minimal if not to be avoided altogether.

Most learners understood the intention for three decision nodes to be required and drew an accurately structured tree. Some learners did not put all labels on branches and/or probabilities on the appropriate branches, meaning that the first accuracy mark was not awarded, but allowed students to score all remaining marks if accurate.

Many learners were able to accurately calculate the end pay-offs and the chance node values. However, a few responses are showing all of their values on another page, which is not an acceptable layout.

Some responses wrote their values in the decision nodes, and 'double-crossed' the inferior options to show the correct path to the decision. As such, the final accuracy mark in (a) was not awarded.

A number of responses gave the correct decision and EMV for part (b); however, some responses could not score this mark as it was dependent on scoring all M marks in (a).

Question 6

This question required learners to use the principle that the flow into a node must equal the flow out, with algebraic flows leading to three simultaneous equations in the unknowns p , q and r . Many responses showed a clear understanding of this requirement and formed appropriate equations. Some responses contained errors in forming or simplifying the equations, while a smaller number contained errors when solving the simultaneous equations. A few responses used synoptic methods, such as matrices, to solve the equations; this was an acceptable approach, although the equations could also be solved directly.

Question 7

This was a typical dynamic programming problem based around maximising the minimum bridge height on a route.

Part (a)

Many responses identified the problem as a maximin problem and applied an appropriate dynamic programming process in part (a).

Part (b)

There were a number of responses who performed an alternative process (minimax, maximising and minimising were all seen on a number of occasions), and a special case was provided which allowed learners to obtain a maximum of 6 marks out of the 11 in (b) for demonstrating accuracy in their chosen process. There were a small number of learners who chose to start at S instead of T; the clear premise of dynamic programming is the idea of working 'backwards' and as such this was also not gaining marks.

Part (c)

Part (c) was a demanding conclusion to the question, which required learners to have found the correct route and height from part (b). Learners were asked to consider whether the blocking of route BD impacted the optimal route; many learners understood that the blocking of the route impacted their solution as BD was on the route, and they automatically defaulted to considering the alternative path BE (which happens, in this case, to be the new optimal route). However, a number of learners understood that with BD blocked there was also the possibility that one of SA or SC could now be optimal instead; a few responses incorrectly cited Bellman's Principle as justification for their partial approach, losing the second B mark.

Question 8

This question focused on a second-order non-homogeneous recurrence relation. Many responses showed a clear understanding of the required method, particularly in forming and solving the auxiliary equation and in identifying the appropriate structure for the solution.

Part (a)

Most learners correctly formed and solved the auxiliary equation, and a majority also understood the structure of the complementary function with repeated roots.

Part (b)

In part (b), many responses recognised the correct form for the particular solution and substituted terms involving $n + 2$, $n + 1$ and n to set up an equation in λ , and many learners found the correct value of λ .

Many learners then found the correct value of λ . Some responses did not continue to the correct general solution, while others contained slips in substitution or rearrangement that affected the final closed form. More successful responses found both the correct closed form and the correct value for the 10th term.

Some responses attempted to determine the constants A and B before considering the particular solution. This generally led to an incomplete general solution and affected the accuracy of the final closed form. Learners should be reminded that, for a non-homogeneous recurrence relation, the complementary function and particular solution need to be considered together before using the given terms to determine the constants.

Summary

- Learners should be reminded of the importance of good basics, such as defining p and $1 - p$ clearly in a game theory question.
- Learners should continue to develop their understanding of recurrence relations.
- Learners should ensure that they are accurate in finding shadow costs and improvement indices and can link their knowledge of degeneracy to finding appropriate improved solutions from the stepping stone method.
- Learners need to understand the requirement to draw appropriate decision trees, including the use of the correct node shapes and the requirement for the nodes to be of sufficient size to hold their working EMVs throughout a decision tree problem.